# Armin Abdollahi

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## Profile Summary

Pioneering **Neuro-AI & BCI Researcher** with a Master's degree in Artificial Intelligence (Perfect Thesis Score: 20/20). Specialized in bridging the gap between human cognition and machines through **Computational Neuroscience** and **Explainable Deep Learning (XAI)**. Focused on developing neurophysiologically robust architectures—such as Convolutional Transformers (CvT)—and the XAI-driven validation of Spectro-Temporal models—to decode motor intent and imagined speech directly from EEG signals. Proven track record in exposing the "Clever Hans" artifact-dependency in SOTA models, driving a paradigm shift toward biologically valid BCI systems. Combining hands-on clinical hardware expertise (fNIRS/EEG) with a published reference book translation, I am seeking a Ph.D. position to design reliable, silent communication interfaces and advance the frontiers of clinical neurorehabilitation.

## Education

**Islamic Azad University, North Tehran Branch** **Sep. 2022 – Graduated Sep. 2025**
*M.Sc. in Artificial Intelligence* *Tehran, Iran*

- **Thesis:** Developing a Transformer-based Deep Learning Model for Signal Decoding in Brain-Computer Interfaces.
- **Defense Score: 20/20 (Perfect Score)** | **GPA:** 3.76/4.00 (18.81/20).
- **Advisors:** Prof. Maliheh Sabeti, Prof. Reza Boostani.
- **Key Achievement:** Demonstrated that high-performing CNNs often overfit to ocular artifacts (EOG). Proposed a novel **Hybrid Transformer-RNN** architecture to ensure neurophysiologically valid decoding.

**Roshdiyeh University** **Sep. 2018 – Jun. 2022**
*B.Sc. in Software Engineering* *Tabriz, Iran*

- Distinguished Student (Ranked top 1% among cohort).

## Research Interests

**Core Areas:** Brain-Computer Interfaces (BCI), Computational Neuroscience, Neural Signal Processing (EEG).

**Methodologies:** Interpretable Deep Learning (XAI), Vision & Convolutional Transformers, Robust Machine Learning.

**Applications:** Neuro-rehabilitation, Silent Speech Interfaces, Assistive Technologies, Medical Image Analysis.

## Publications & Presentations

### Journal Publications

- **A. Abdollahi**, M. Sabeti, R. Boostani. "The Illusion of Accuracy in Rehabilitation BCI: High-Performance CNNs Learn Artifacts, Not Neural Intent." *Biomedical Signal Processing and Control (BSPC)*. (Under Review).
- **A. Abdollahi**, M. Sabeti, R. Boostani. "A Robust Benchmark for Rehabilitation BCI: Validating Convolutional Transformers on Neurophysiologically Isolated EEG Signals." *Biomedical Signal Processing and Control (BSPC)*. (Under Review).
- S. Kasaiyan, **A. Abdollahi**, et al. "Deep Learning for Imagined Speech Classification: A Review and New Architecture." (In Preparation – Target Journal: IEEE Transactions on Neural Systems and Rehabilitation Engineering).

### Books & Translations

- Alexander Guger. *"Brain-Computer Interface: A State-of-the-Art Summary 12"*. Persian Translation by **A. Abdollahi**. (Published, Jan. 2026).

### Conference Proceedings & Presentations

- **A. Abdollahi**, M. Sabeti, R. Boostani. "Bone Age Estimation Through Hand X-Ray Analysis with Visual Transformer Model." *The International Conference on Medicine and Artificial Intelligence in Health Improvement (IOAS)*. Germany, Aug 2025. [**Poster Presentation, Published in Conference Proceedings**]
- **A. Abdollahi**, M. Sabeti, R. Boostani. "Bone Age Estimation Through Hand X-Ray Analysis with Visual Transformer Model." *2nd International Congress on AI in Medical Sciences (AIMS)*. Tehran, Iran, May 2025. [**Poster Presentation**]

## Research Experience

**Neurorehabilitation & BCI Research** | ***Explainable Deep Learning*** *Tehran, Iran*

- Developed a comparative benchmark (**NeuroValid**) using Optuna to evaluate CNNs (EEGNet) vs. Vision and Convolutional Transformers (ViT, CvT) across isolated EEG frequency bands.
- Utilized **Grad-CAM**, **Saliency Maps**, and **Representational Similarity Analysis (RSA)** to uncover the "Clever Hans" effect, proving SOTA CNNs classify based on ocular artifacts (EOG) rather than true neural intent.
- Engineered a robust **Convolutional Transformer (CvT)** architecture that resists artifact interference and maintains high performance (**80.78% F1-Score**) on strictly filtered (Mu-band) data, where standard CNNs suffer significant performance drops.

**Silent Communication BCI** | ***Imagined Speech Decoding (Joint Project)*** *Current*

- **Signal Processing & Artifact Rejection:** Engineered the critical preprocessing and artifact mitigation pipelines (ICA, ASR) to isolate pure neural correlates from severe facial EMG contamination during speech tasks.
- **XAI & Biological Validation:** Evaluated the neurophysiological fidelity of a co-developed **Spectro-Temporal Transformer** architecture using Explainable AI (SHAP), ensuring the model decodes true linguistic intent rather than articulation noise.
- **Data Pipeline Engineering:** Developed the end-to-end data handling and feature extraction pipeline for the **KARA ONE** dataset, enabling robust classification of phonemes and words for assistive devices.

**National Brain Mapping Laboratory (NBML)** | ***Research Intern (fNIRS & EEG)*** *Tehran, Iran*

- Acquired hands-on expertise in multimodal neural signal recording, specifically utilizing **fNIRS** and **EEG** modalities for neuro-cognitive assessments.
- Designed and executed experimental paradigms for neurofeedback and biological signal analysis, bridging the gap between hardware acquisition and computational processing.

**Medical Imaging Project** | ***Deep Learning Applications*** *[GitHub]*

- **Bone Age Estimation:** Implemented a Vision Transformer (ViT) for regression on hand X-ray images (Atlas dataset), achieving an **MAE reduction of 12%** compared to traditional CNN baselines.

## Professional Experience

**AI Specialist & Bio-Signal Analyst** **Current**
*White Tree Intl* *Tehran, Iran*

- Applying cutting-edge computational techniques and deep learning models to real-world neurotechnology and mental health challenges.
- Analyzing biological signals with a specific focus on pediatric neuropsychology applications.

## Teaching Experience

**Head Teaching Assistant (Promoted in Fall 2025)** **Sep 2023 – Present**
*Islamic Azad University, North Tehran Branch* *Tehran, Iran*

- **Deep Learning:** Managed a TA team for 70+ students. Designed advanced course materials focusing on **Transformers**, **CNNs**, and **Optuna**; conducted hands-on live coding sessions.
- **Digital Signal & Image Processing:** Instructed 50–90 students per semester across 6 terms. Taught practical EEG analysis (**MNE-Python, EEGLAB**) and Computer Vision architectures (**U-Net**). Created specialized **[Video Tutorials]**.
- **Machine Learning:** Assisted Prof. Sabeti by independently curating practical datasets, designing evaluation metrics, and providing strategic mentorship for ML term projects.

## Academic Leadership & Seminars

- **Workshop Instructor: BCI & Brain Signal Processing** (Feb 2026). Organized and led specialized hands-on workshops for students, focusing on practical EEG data analysis, artifact removal, and preprocessing pipelines using MNE-Python and EEGLAB.
- **Webinar Co-Organizer & Technical Lead: Attention in Deep Learning Models** (Jun 2024). Co-hosted a technical webinar with Prof. Sabeti. While she provided theoretical foundations, I independently led the **live coding sessions**, demonstrating the practical integration of attention mechanisms in NLP and Vision tasks **[Watch Session]**.
- **Tutorial Lead: Applied Deep Learning in Bio-Signals** (2024). Conducted technical lab sessions demonstrating the implementation of deep learning models, including Sleep Apnea event detection using 1D-CNN and Transformer architectures.

## Honors & Awards

- **Defended Master's Thesis with Highest Honors (20/20):** Islamic Azad University, North Tehran Branch (2025).
- **Letter of Appreciation for Guest Speaker:** Recognized for presenting "Attention Mechanisms in Deep Learning Models", Islamic Azad University (Jun 2024).
- **Academic Scholarship Recipient:** Awarded for three consecutive semesters during M.Sc. studies for maintaining high academic standards (2022 – 2024).
- **Outstanding Student Recognition:** Ranked in the top 1% of the B.Sc. Software Engineering cohort at Roshdiyeh University (2022).

## Technical Skills

**Programming:** Python (NumPy, Pandas, Matplotlib), MATLAB, C++, SQL.

**Machine & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Optuna, Hugging Face (Transformers).

**Explainable AI (XAI):** Grad-CAM, Saliency Maps, Representational Similarity Analysis (RSA), SHAP.

**Signal Processing: MNE-Python**, EEGLAB, BioSig, Wavelet Transform, CSP, ICA.

**Tools & Platforms:** Git, GitHub, Linux (Ubuntu), Docker, LaTeX.

**Languages:** Persian (Native), Azerbaijani/Turkish (Fluent), English (Professional Proficiency, IELTS: 6.5), German (A1).

## Certificates & Specialized Training

**BCI & Neurotechnology Spring School** (140 hours) – g.tec medical engineering *May 2024*

**Advanced Neuroimaging & Data Processing Series** (Combined 100+ hours) – National Brain Mapping Lab (NBML) *2023 – 2024*

**Machine Learning Algorithms in EEG-Based BCI** (12 hours) – Connectome Academy *Jan 2024*

**Neuro-AI Program & Winter Seminar Series** – Shahid Beheshti & Sharif University *2023 – 2024*

## References

*Reference letters are available upon request.*


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